Samuel Habib

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Technical Skills

Languages: Python, Ruby, C, C++, C#, Java, JavaScript, Typescript, Perl, PHP, SQL, HTML/CSS, HTML5, XML, Groovy
Frameworks: AWS, Nodejs, Reactjs, Flask, Unity, Docker, MATLAB, SocketIO, Bootstrap, Git/Github, JIRA, VS code, Visual Studio, Confluence, SSH, Vim, SCP, Bash, Linux, Google Cloud Platform, Arduino, Jenkins, JQuery, NLP, IntelliJ

Experience

Apple Sept 2023 – Present
Software Development Engineer *Austin, TX, USA*

- Contributed to **Apple Travel**, enhancing approval workflows for purchases, travel, and access across Apple employees and vendors.
- Led development of an **AI-based** receipt parsing service for Apple Travel wallet, combining **LLM extraction**, rule-based validation, and schema normalization to reliably process third-party receipts and provide end-to-end expense automation.
- Led **receipt integrations** in **Apple Travel**, exposing APIs to third-party partners, setting up and maintaining **SSL/TLS certificates**, coordinating testing, and automating expense reconciliation and bookings, reducing manual processing by **80%**.
- Designed scalable **RESTful APIs** with **Java**, **Spring Boot**, and **WebFlux**, enabling high-throughput and low-latency processing.
- Owned and developed **BART** system, replacing spreadsheets with a structured budget reconciliation system, improving financial tracking and reducing errors by **95%**, while authoring **technical documentation** to enhance team onboarding and maintainability.
- Architected **OIDC authentication** with **RBAC**, ensuring secure role-based access and seamless system integration.
- Optimized complex **Oracle SQL** queries, reducing data retrieval time by **33%** for budget tracking and expense forecasting.
- Implemented **Solr-based indexed search**, reducing approval lookup times by **90%** and improving auditability and compliance.
- Achieved **85%+ unit test** coverage with **JUnit** and **Mockito**, improving system reliability and reducing defects by **20%**.

Amazon June 2024 – Sept 2024
Software Development Engineer Intern *Toronto, ON, CA*

- Developed an internal **visualization** tool for **FBA** sellers, enabling efficient analysis of **business metrics** and insights.
- Utilized **CDK** to deploy an **S3**, **SQS**, **Lambda** flow for storage, querying, and retrieval across distributed environments.
- Created a **data** system to asynchronously pull and convert **S3** metrics into **Java** data models for real-time investigation.
- Built a **scalable** pipeline using **S3** notifications, **SQS**, and **Lambda**, processing **70,000+** requests per hour across regions
- Modified **React.js** components for multi-tabbed data visualization, supporting exception handling and fluid interaction
- Improved **JSON** parsing with **Java** and **Lombok**, designing flexible models, **interfaces**, and ensuring maintainability.
- Achieved **83%** test coverage by creating **55 JUnit tests**, ensuring high standards, reliability and robust functionality.

Amazon May 2023 – Sept 2023
Software Development Engineer Intern *Toronto, ON, CA*

- Designed and built a comprehensive end-to-end project, leveraging **AWS** services **SQS, DynamoDB, DAX, SNS, VPC**
- Utilizing **Kotlin** developed a partitioning mechanism to distinguish between data types, enhancing data accuracy by **90%**.
- Effectively managed data storage, implementing **DAX caching** strategies that reduced costs by an impressive **45%**.
- Managed the entire **project cycle**, integrating it into the team's system, ensuring no latency, and achieving **zero** delays
- Established **CloudWatch** alarms and custom real-time **dashboards** to monitor **p90**, **max**, **min** and **average metrics**
- Designed **CDK infrastructure** to be easily configurable for future datasets, ensuring **scalability** and **adaptability**.
- Introduced the capability of handling over **2 billion** records per hour in the largest dataset using **AWS** services.

AMD (Advanced Micro Devices) May 2022 – May 2023
Software Engineer Full Stack Intern *Toronto, ON, CA*

- Built auto-scaling and self-healing back-end using **Ruby** & **YAML** performing **Regression** tests of **1k+** tests in parallel.
- Leveraging **Ruby** to develop services boosting work efficiency by **70%** through automation of **CI/CD** logging **pipelines**.
- Creating a gateway for parsing and filtering log files using **REGEX**, further standardizing the current **logging** dashboard
- Implementing Agile methodology's best practices that increase software delivery processes' speed by **50%**.
- Developing and maintaining **PHP** front-end code relaying data to **MySQL** back-end utilizing **REST API** requests.
- Utilizing **JIRA** and communicating with other engineers to monitor and resolve various issues in the back-end.

Projects

Mental Health Crisis Detection AI Chatbot | *Python, Flask, React, GPT-3.5 Turbo, RoBERTa*

- Designed a responsive AI chatbot with **Flask** and **React**, integrating **openAI** GPT-3.5 for personalized user interactions.
- Implemented a **RoBERTa** model for precise sentiment analysis, adapting responses based on users' emotional states.
- Developed a robust system with **NLP** algorithms to detect distress signals and promptly alert emergencies.
- Integrated **Firebase DB** for secure and efficient data storage and user authentication, fortifying the chatbot's reliability.

Education

Toronto Metropolitan University Toronto, ON
B.Eng Computer Engineering, CGPA: 3.90 *Aug 2019 - May 2024*
Relevant Courses: Digital Systems • Software Systems • Data Structures and Algorithms • Object-Oriented Analysis and Design • Database Systems I • Microprocessor Systems • Signals and Systems I • Computer Visions • Distributed Systems